

Name of Experiment: Spatial Resolution Reduction of a Grayscale Image.

Introduction: In this experiment, a 512×512 grayscale image was used to study the effect of reducing spatial resolution. The image resolution was decreased by half at each step (512, 256, 128, 64, 32, and 16), and then each image was resized back to the same display size for comparison. OpenCV was used for image resizing and Matplotlib was used for visualization. The purpose of this experiment was to observe how image quality changes as spatial resolution decreases.

Code:

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread('pepe.PNG', 0)
img = cv2.resize(img, (512, 512))
resolutions = [512, 256, 128, 64, 32, 16]

plt.figure(figsize=(18, 6))

for i, res in enumerate(resolutions):
    small_img = cv2.resize(
        img,
        (res, res),
        interpolation=cv2.INTER_AREA
    )

    display_img = cv2.resize(
        small_img,
        (512, 512),
        interpolation=cv2.INTER_NEAREST
    )

    plt.subplot(1, 6, i + 1)
    plt.imshow(display_img, cmap='gray')
    plt.title(f'{res}x{res}')
    plt.axis('off')

plt.tight_layout()
plt.show()
```

Input Image:



Output Image:



Discussion: The output shows six versions of the same grayscale image with different spatial resolutions.

- **512×512 Resolution:**
The original image appears sharp and clear with all details visible.
- **256×256 Resolution:**
Slight loss of detail is noticeable, but the image still looks relatively smooth and recognizable.
- **128×128 Resolution:**
The image starts losing fine textures and edge sharpness. Some pixelation becomes visible.
- **64×64 Resolution:**
Pixel blocks become more noticeable, and image quality decreases significantly.
- **32×32 Resolution:**
The image appears blurry and heavily pixelated. Most fine details are lost.
- **16×16 Resolution:**
The image becomes extremely distorted and blocky, making object recognition difficult.

Reducing spatial resolution decreases image quality and removes visual details. As the resolution becomes smaller, the image appears more pixelated and unclear. This experiment shows the importance of spatial resolution in digital image processing.